



UNIVERSITY OF MANITOBA
Department of Mathematics

COURSE: *Math 1210*
DATE: *June 1, 2023*
EXAM: *Term test 3*

EXAMINER: *Romain Branchereau*
TIME: *50 Minutes*

First name (please write as legibly as possible within the boxes)

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Last name

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Student ID number

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Grading table

Question:	1	2	3	4	Total
Points:	12	6	15	13	46
Score:					

Instructions

- Answer all questions on the exam paper in the space provided beneath the question. Unless otherwise specified, your answer must be justified. Unjustified answers will receive little or no credit.
- If you need more space, continue on the back of the same page, clearly indicating that your work is to be continued.
- The value of each question is indicated beside the statement of the question. The total value of all questions is 46 points.
- No texts, notes, or other aids are permitted. There are no calculators, cellphones, or any other electronic device permitted.
- Do not remove staples or any pages from the exam except the last blank page.
- If you have any questions, feel free to ask me.
- Good Luck!



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1. (12 points) Consider the matrices

$$A = \begin{pmatrix} 2 & -1 \\ 0 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & -5 \end{pmatrix} \quad C = \begin{pmatrix} 1 & -2 \\ 2 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- (a) Evaluate the expression $B^T(A - 2C)$ if it is defined, or explain why it is undefined.

- (b) Evaluate the expression $A^T C - B^2$ if it is defined, or explain why it is undefined.

- (c) Evaluate the expression $2AC - D^T B$ if it is defined, or explain why it is undefined.

- (d) Suppose that X is a matrix such that the expression $(BX)^T$ is a well-defined 5×2 matrix. What is the size of X ?



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2. Consider the vector $\mathbf{u} = \langle 2m, -3, 1 \rangle$ and $\mathbf{v} = \langle -m, m, 3 \rangle$ where m is a real number.

(a) (3 points) For which value(s) m are the two vectors $\mathbf{u}$ and $\mathbf{v}$ perpendicular?

(b) (3 points) For $m = -1$, find a vector of length 3 in the opposite direction to $\mathbf{u}$.



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3. *The three parts (a),(b),(c) are independant. Part (c) is on the next page.*

- (a) (5 points) Find parametric and symmetric equations for the line through the points $P(2, -1, 3)$ and $Q(3, 3, 1)$. Use the symmetric equation to find two planes that intersect along this line.

- (b) (5 points) Find an equation of the plane through the point $R(5, 2, -2)$, and perpendicular to the planes $x + 3y - 2z = 11$ and $2x - y = 23$.



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- (c) (5 points) Find the intersection between the line $x = 2 + t$, $y = -1 + 4t$, $z = 3 - 2t$ and the plane $x + 2y + 3z = 3$.



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4. Consider the system

$$\begin{aligned}2x_2 + 7x_4 &= 0 \\x_2 - x_3 + 2x_4 &= 0 \\x_1 + 2x_3 &= 0\end{aligned}$$

- (a) (3 points) Without solving the system, explain why it cannot have a unique solution.
- (b) (10 points) Use Gauss-Jordan elimination to find the solutions of the system. Indicate the rank of the system, the number of parameters of the system and list the basic solution(s) if they exist.



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